# Unveiling the Unknown: A SAM Guided Open World Object Detection Method for Remote Sensing

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***Abstract*—Despite the remarkable success of remote sensing object detection, these methods primarily operate under a closed-world paradigm, as they often misclassify or ignore novel objects in real-world scenarios. To address this limitation, open-world object detection (OWOD) has emerged, enabling the discovery and incremental learning of new categories. However, existing OWOD approaches typically distinguish between known and unknown categories based on feature distance, overlooking the inherent challenge in remote sensing: large intraclass variation versus small interclass variation. To bridge the gap, we propose a novel framework for SAM-guided OWOD tailored for remote sensing imagery. Our approach is designed to leverage SAM's capabilities for discovering new categories while systematically handling the noisy labels produced by SAM. We introduce four key components: 1) a multiscale feature fusion perception (MFFP) module to enhance the detection of unknown objects across various scales in remote sensing; 2) a cross-layer cascaded decoupling decoder (CCDD) to alleviate the optimization conflicts between objectness and classification tasks for similar known and unknown classes in remote sensing images; 3) a label mapping alignment (LMA) mechanism to adaptively filter background noisy proposals from SAM; and 4) an active learning (AL) strategy is proposed to intelligently select exemplars for robust incremental learning. Extensive experiments on benchmark remote sensing datasets, including DIOR, DOTA, and NWPU, demonstrate that our method significantly improves the recall of unknown objects while maintaining robust detection performance for known classes, demonstrating the effectiveness and potential of the SAM-guided OWOD paradigm.**

***Index Terms*—Incremental learning, object detection, open-world object detection (OWOD), remote sensing, segment anything model.**

## I. Introduction

With the rapid advancement of remote sensing technology, high-quality aerial and satellite imagery has become increasingly abundant, making data-driven deep learning methods the cornerstone of object detection [1], [2]. From early convolutional architectures like Faster R-CNN [3], [4], [5] and YOLO [6], [7], [8] to modern transformer-based models like DETR [9], [10], [11], object detection in remote sensing has achieved remarkable success in identifying predefined targets such as vehicles, ships, and buildings. However, these methods primarily operate under a closed-world paradigm, where training and inference are confined to a fixed set of known classes. This limitation severely hinders their practical application in real-world scenarios, where models frequently encounter novel or unannotated objects that are consequently ignored or misclassified as background.

To bridge this gap, open-world object detection (OWOD) [12] has emerged as a more demanding and realistic task, aiming to mirror the human ability to explore and learn from a dynamic environment. OWOD requires a model not only to detect and classify known objects accurately but also to identify previously unseen objects as "unknown" and incrementally learn these new classes over time without catastrophic forgetting when new labels are provided. This capability is crucial for dynamic remote sensing applications, including maritime surveillance, disaster response, and military reconnaissance, where the environment is constantly evolving.

A critical distinction must be made between OWOD and the related paradigm of open vocabulary object detection (OVOD). Although both frameworks address the challenge of novel categories, their core mechanisms differ fundamentally. OWOD operates without any prior semantic information about unknown objects, focusing on first discovering them as generic unknown instances and then incrementally learning their specific classes as labels become available. In contrast, OVOD leverages large-scale vision-language models to detect objects described by arbitrary text queries at inference time, thereby eliminating the need for an incremental learning phase. This work is situated within the OWOD paradigm, which is particularly well-suited for remote sensing applications where novel objects often emerge unexpectedly and must be identified before linguistic descriptions can be formulated. The core challenge in OWOD is to distinguish unknown objects from the background without direct supervision. To address this, existing methods have explored various strategies to learn a general objectness model. For instance, ORE [12], a pioneering work, employs an energy-based classifier and

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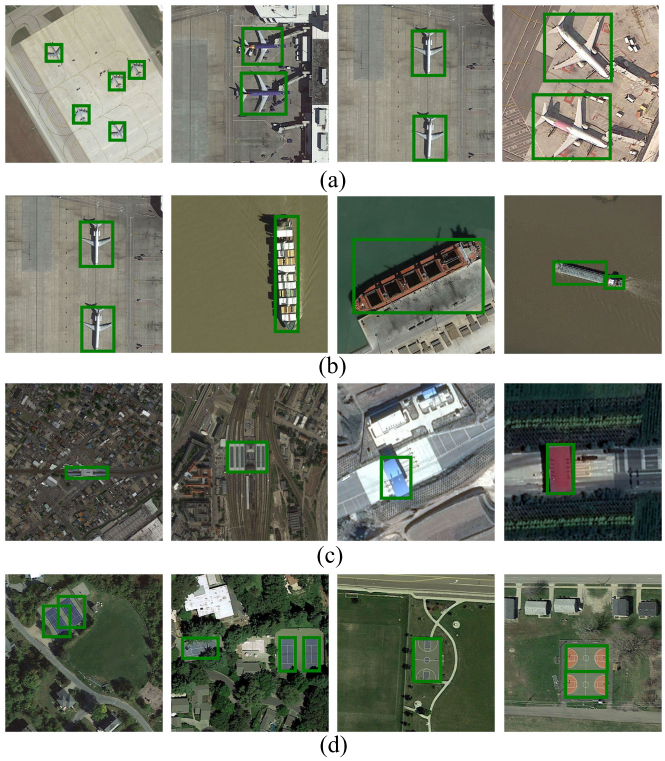



Fig. 1. Characteristics of remote sensing images. (a) Airplane objects at different scales. (b) Ship objects at different scales. (c) Comparison of (first two columns) ETs versus (last two columns) TSs. (d) Comparison of (first two columns) TCs versus (last two columns) BCs.

contrastive clustering to separate known objects from potential unknowns. Transformer-based methods like OW-DETR [13] utilize a pseudo-labeling strategy, selecting regions with high feature activation as unknown object candidates for supervision. More recently, PROB [14] introduced a probabilistic approach, modeling the feature space as a class-agnostic Gaussian distribution to identify outliers as unknowns. Specifically for remote sensing, OPODet [15] tackles the unique challenges of arbitrary object orientation and large feature disparities by introducing an oriented unknown-aware region proposal network. Despite these efforts, these methods typically learn from a relatively small set of known classes, which limits their ability to acquire the comprehensive, generic knowledge needed to perceive the vast diversity of objects in the open world.

Recent advancements in large vision models (LVMs) [16], [17], [18], [19], [20], [21], trained on unprecedented scales of data, offer a promising opportunity to fundamentally advance OWOD by harnessing their rich, generic knowledge. The segment anything model (SAM) [16], in particular, is an excellent candidate due to its exceptional ability to perform class-agnostic segmentation with high-quality localization, which aligns perfectly with the goal of discovering unknown objects. By leveraging the powerful prior knowledge embedded in SAM, we can move beyond learning objectness from a small set of known examples and instead tap into a more universal understanding of what constitutes an object. However, the

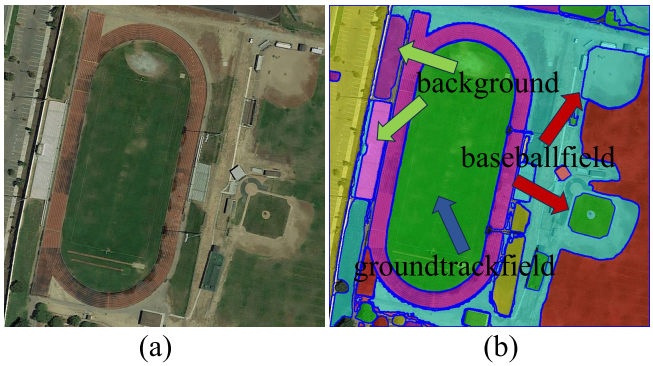



Fig. 2. SAM segmentation of the entire image not only provides known (blue arrow) and unknown classes (red arrows) but also generates background noises (green arrows). (a) Original image. (b) SAM segmentation results.

OWOD method that directly uses SAM as an unknown class prior in remote sensing still has challenges.

### A. Unique Characteristics of Remote Sensing Imagery

Remote sensing imagery presents two fundamental, compounding challenges for OWOD. On the one hand, extreme intraclass scale variation means that objects within the same category differ dramatically in size, such as large passenger aircraft versus small planes [see Fig. 1(a)] or large cargo vessels versus small boats [see Fig. 1(b)]. This vast scale range hinders a model's ability to generalize and complicates the discovery of novel objects across different scales. On the other hand, low interclass variance causes distinct categories to exhibit high visual similarity; for instance, expressway toll stations (ETs) like train stations (TSs) [see Fig. 1(c)], or tennis courts (TCs) like basketball courts (BCs) [see Fig. 1(d)], can appear as similar rectangular structures. This ambiguity increases the risk of misidentifying unknown objects as known classes, forcing representation adjustments that degrade performance on previously learned categories. Together, these characteristics create a uniquely challenging scenario where models must simultaneously manage massive scale variations while distinguishing subtle interclass differences, rendering the direct application of existing OWOD methods inadequate.

### B. Noise SAM Labels

When leveraging SAM to generate pseudo-labels for potential unknown classes in remote sensing, a significant challenge arises: the creation of background noise labels. Because SAM is class-agnostic, it will segment not only genuine novel objects but also many nonsemantic regions like shadows, patches of bare earth, or road fragments, treating them all as potential unknowns as shown in Fig. 2. Therefore, it becomes critically important to establish a robust mechanism to differentiate among the generated masks, effectively categorizing them into three distinct groups: known classes, true unknown classes, and background. This careful separation is essential to prevent polluting the training data with false positives and to ensure that the model can accurately expand its knowledge without degrading its performance on existing categories.

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To address these critical challenges in remote sensing image analysis, we propose an SAM-guided OWOD method comprising four key components. First, to handle extreme intraclass scale variation, we introduce a multiscale feature fusion perception (MFFP) module that aggregates contextual information across network layers for robust multiscale object perception. Second, to prevent confusion between visually similar known and unknown classes, we employ a cross-layer cascaded decoupling decoder (CCDD) that separates localization and classification pathways, ensuring new class discovery does not degrade existing accuracy. Third, to filter noisy background segments from SAM, we utilize a label mapping alignment (LMA) mechanism that adapts pseudo-labels by jointly considering objectness confidence and localization consistency. Finally, to intelligently select exemplars for incremental learning while minimizing annotation costs, we propose an active learning (AL) strategy that mines hard negative and prototypical positive samples based on objectness scores, stabilizing old knowledge while preparing the model for novel categories. Our method and closed-set object detection can be illustrated as Fig. 3, and our contributions can be summarized as follows.

1) An MFFP module is introduced to enhance the model's ability to detect unknown objects across various spatial resolutions in remote sensing scenes by systematically fusing high-resolution spatial details from shallow layers with rich semantic context from deep layers.
2) A CCDD is designed to mitigate the optimization conflict between objectness estimation and classification, which avoids confusion between known and unknown classes by physically separating these two tasks into distinct but coordinated learning branches.
3) An LMA mechanism is developed to effectively filter the noisy labels generated by SAM, reducing error propagation and preserving detection precision on known classes by using the correlation function between predicted results and SAM results.
4) An AL strategy is employed to intelligently select informative samples for incremental learning, to solve the high-cost problem of manual annotation in incremental learning by using simple and difficult mixed samples for model prediction.

Extensive experiments on benchmark remote sensing datasets, including DIOR [22], DOTA [23], and NWPU [24], demonstrate that our method significantly improves the recall of unknown objects while maintaining robust detection performance for known classes.

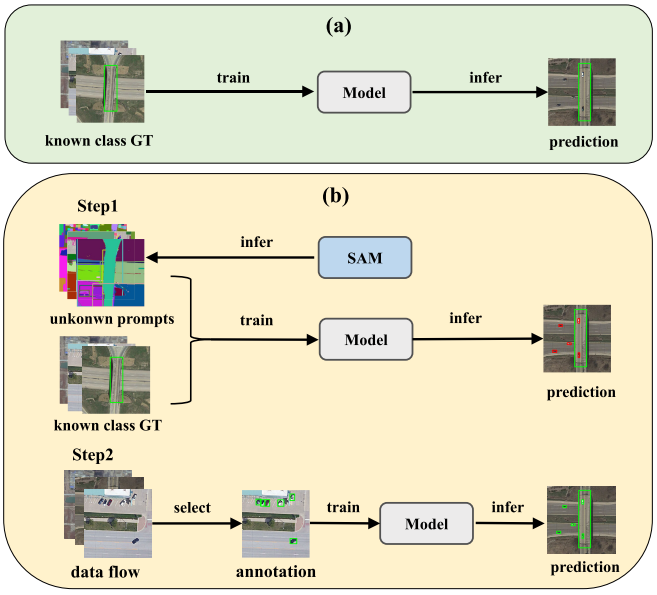



Fig. 3. Illustration of closed-set object detection and our method. (a) Closed-set object detection can only detect known classes. (b) Our method can actively discover unknown classes in Step 1 and can continue to detect unknown classes by actively learning to increment the interested unknown classes with a small amount of data in Step 2.

## II. Related Works

### A. Object Detection

In traditional remote sensing research, object detection has primarily followed a closed-world paradigm, where both training and inference rely strictly on a fixed set of object classes. Early works adapted convolutional architectures, such as R-CNN, Fast R-CNN, Faster R-CNN, SSD, YOLO, and RetinaNet to remote sensing imagery [25], [26], [27], [28], [29], [30], demonstrate that our method significantly improves the recall of, achieving impressive accuracy in detecting vehicles, buildings, and ships. However, these methods still relied on extensive anchor design, nonmaximum suppression, and ad hoc heuristics. Subsequently, anchor-free methods [31], [32], [33], [34], [35] emerged with the assertion of not using manually designed anchor boxes and have further improved the detection performance.

Inspired by natural image detection, the DETR introduced a paradigm shift by treating object detection as a sequence-to-set problem. DETR uses a transformer encoder–decoder and bipartite matching loss to remove manual components like anchor boxes and NMS, enabling an end-to-end trainable framework. Although original DETR struggled with convergence time and small object detection, later variants such as Deformable-DETR featuring a deformable attention module significantly improved its performance and training efficiency, making DETR-like architectures practical in remote sensing domains [36], [37]. However, all these methods focus on detecting labeled objects in a closed set and are unable to predict potential objects without ground truth labels. In contrast, our method predicts objects from both labeled known classes and unlabeled unknown classes in remote sensing images.

### B. Open-World Object Detection

OWOD, introduced to overcome the constraints of fixed category and closed training paradigms, has lately drawn widespread interest because it suits unconstrained, continually changing environments. OWOD is required to recognize both known and unknown objects, as well as to incrementally learn newly introduced knowledge. To learn knowledge incrementally, a continual learning paradigm is adopted, where new data and novel classes are introduced as time series, such as

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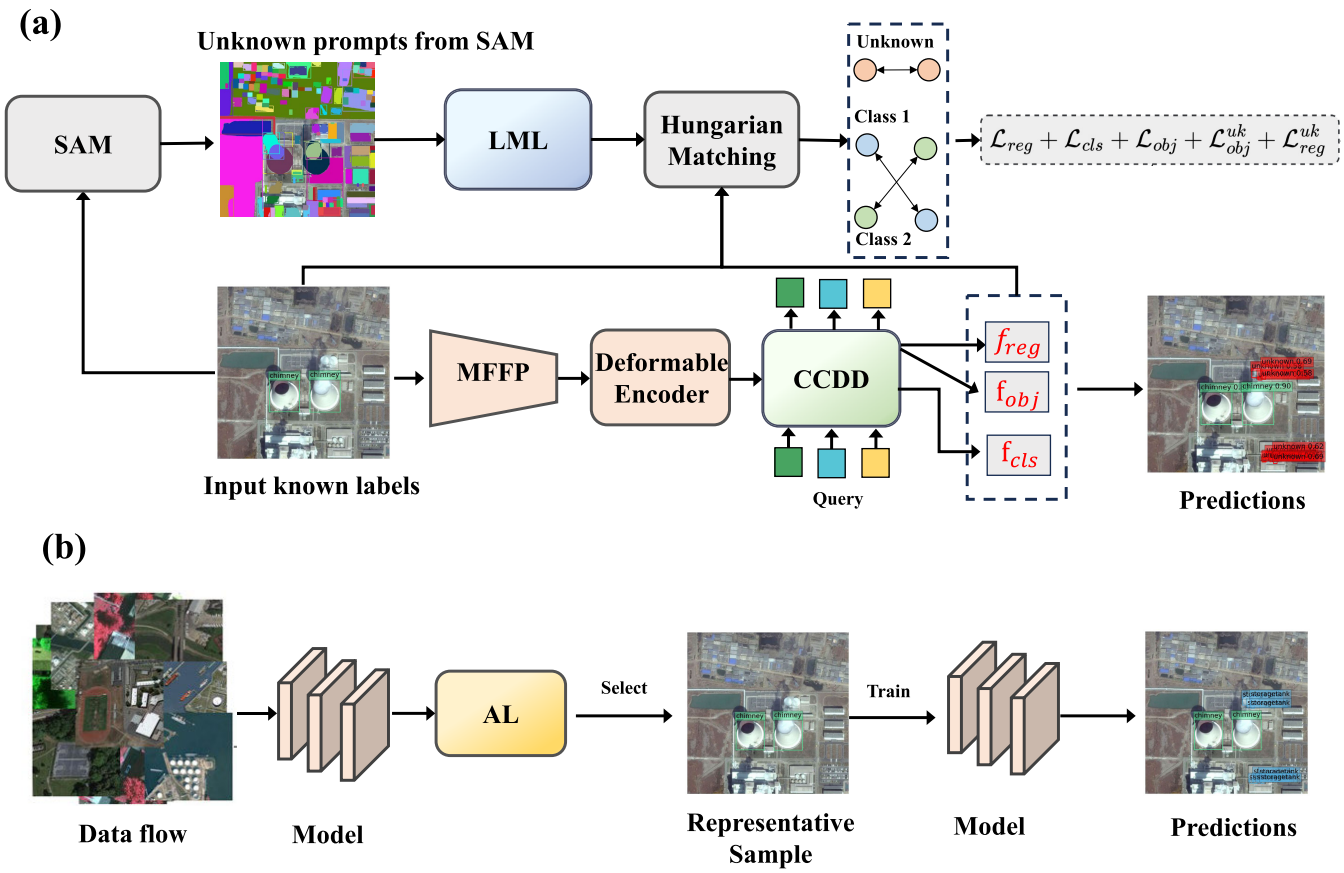



Fig. 4. Overall framework of our method. We adopt D-DETR as the basic detector and integrate an objectness layer to recognize unknown objects. Then, we have access to two types of labels, including accurate known object labels and noisy unknown object labels generated by SAM. (a) In Step 1, our method contains: 1) MFFP module; 2) CCDD module; and 3) LMA module for discovering unknown categories. (b) In Step 2, we propose 4) the AL strategies for learning unknown labels.

the incremental object detection task [38], [39], [40], [41], [42], [43], [44].

Pioneered by Joseph et al. [12], OWOD frameworks tackle the dual task of recognizing known classes and discovering unknown ones, followed by incremental learning when labels become available. ORE integrates contrastive clustering, an energy-based unknown identification head, and an unknown-aware proposal network to identify novel objects and continuously update the detector. OW-DETR [13] introduces attention-driven pseudo labeling, novelty classification, and objectness scoring into the DETR architecture. The OW-DETR enhanced the performance using multiscale self-attention and deformable receptive fields. In addition, Zhao et al. [45] extended the OWOD framework with an auxiliary proposal advisor and a class-specific expelling classifier. Dong et al. [46] extended the OWOD framework with an auxiliary proposal advisor and developed a two-stage training approach based on Deformable-DETR to mitigate catastrophic forgetting of unknown class annotations. Yu et al. [47] proposed an efficient OWOD solution using discriminative class prototype learning.

A notable advancement in open-world object detection for remote sensing is OPODet. Recognizing the challenges of arbitrary object orientations and large interclass feature discrepancies in aerial imagery, OPODet integrates a novel oriented unknown-aware region proposal network into remote sensing object detectors. Saini et al. [48] propose a multimodal pipeline that integrates threshold-based region proposals for unknown objects, followed by semantic labeling using multimodal large language model. OWOD algorithms closely mirror real-world conditions and human learning processes by enabling models to discover unknown objects and continuously learn with minimal labeled data. Integrating OWOD into remote sensing imagery is thus essential. However, current limitations hinder their effective application and lack a focus on continual detection capabilities in this domain, making it a challenge that our proposed method seeks to address.

### C. Segment Anything Model

SAM is a foundational visual model that can perform zero-shot segmentation for any object in an image using simple prompts like points or boxes. In remote sensing, it is widely applied for the rapid and accurate extraction of various geospatial features, such as buildings, water bodies, and agricultural fields, significantly accelerating tasks like land cover mapping and object detection [49], [50], [51], [52], [53]. SAM operates by generating masks from a dense grid of sampling points distributed across the image, enabling it

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to segment a broad spectrum of objects without relying on class labels. This class-agnostic nature aligns well with the goals of OWOD, particularly in detecting unknown objects. Moreover, SAM's precise localization ability further enhances its suitability for OWOD. Therefore, harnessing the rich, generalizable knowledge embedded in publicly available models presents significant opportunities for improving performance in downstream tasks.

## III. Proposed Method

### A. Problem Description and Network Architecture

In general object detection, a set of known classes $K = \{1, 2, 3, \ldots, C\}$ is given, along with the corresponding training set $D = \{I, Y\}$ consisting of $N$ images $I = \{I_1, I_2, \ldots, I_N\}$ and their corresponding labels $Y = \{Y_1, Y_2, \ldots, Y_N\}$. Each label of Y contains m annotations $Y_i = \{Y_1, Y_2, \ldots, Y_m\}$, each of which contains classification label $l_j \in K$ and box label $b_j = \{x_j, y_j, w_j, h_j\}$, where $(x_j, y_j)$ is the top-left point of the box and $w_j$ and $h_j$ are the width and height of the box, respectively. General object detection methods are unable to detect these potential unknown object classes U that are not labeled.

Open-world detection is motivated by the rigidity of fixed set training and the missed unknown objects that plague standard detectors. OWOD typically involves a set of sequential tasks for incremental learning. Given a specific time or task t, we have known classes denoted as $K^t$, and an associated dataset $D^t = \{I^t, Y^t\}$, which consists of $N^t$ images It and their corresponding labels $Y^t$. Unlike general object detection, the OWOD model $M^t$ predicts not only the known objects with $K^t$ classes but also the unknown objects with an additional class $(K^t + 1)$. Then, these unknown objects are sent to an oracle to label new objects of interest with $U^t$ classes, producing $D^{t+1}$. At the next time, given only $D^{t+1}$, $M^t$, and a set of memorized training data for exemplar replay to produce $M^{t+1}$ that can detect $K^{t+1} = \{K^t \cup U^t\}$ known classes. This is a multiround human–AI interaction process, proving particularly practical for open-world environments. However, owing to the limited availability of object labels in OWOD, achieving comprehensive detection of unknown objects presents a considerable challenge.

SAM-guided OWOD, as we proposed in this work, leverages the power of VLMs to enhance the detection of unknown objects, enabling safer and more robust object detection in diverse real-world scenarios. Compared to OWOD, SAM-guided OWOD adds additional unknown object labels generated by the SAM model. SAM densely samples an $n \times n$ lattice of foreground points; every point spawns a mask that may delineate a legitimate object, tiling the whole image with candidate regions. The resulting bounding boxes $B_{\text{sam}}$ are then derived from the smallest enclosing rectangle of each mask. $B_{\text{sam}}$ is class-agnostic and serves as an auxiliary supervision for unknown object detection, which significantly enhances the capability of recalling more unknown objects. However, as has been claimed, the existing noise in $B_{\text{sam}}$ is a nonnegligible problem in SAM-guided OWOD.

Fig. 4 illustrates the framework of our method. We build upon D-DETR as our foundational detector, incorporating a backbone, a deformable encoder, and decoder into the architecture. The D-DETR model generates query embeddings, which are then fed into the objectness, classification, and regression heads. The classification probability and the objectness probability are multiplied to get the final predictions, including known and unknown object classifications. We explicitly illustrate the placement of the proposed methods: 1) the MFFP module; 2) the CCDD module; 3) the LMA module; and 4) the AL module.

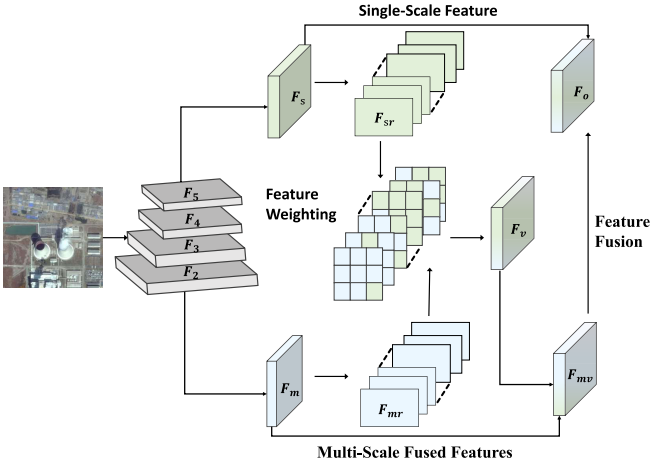



Fig. 5. Illustration of MFFP. By using multiscale features to make up for single-scale features, the model can obtain more representative features and can better deal with the problem of discovering unknown labels.

In the following, we first introduce an MFFP module to improve the model's ability to perceive unknown classes. Next, a CCDD is designed to mitigate the optimization conflict between objectness and classification layers, improving the detection accuracy of known objects. In addition, an LMA mechanism is proposed to adaptively filter out noisy labels generated from SAM, establishing robust learning for noisy labels and preventing the uncontrolled expansion of objectness semantics, mitigating the erroneous classification of unknown objects as known objects. Finally, we introduce the AL method to enhance the incremental process.

### B. Multiscale Feature Fusion Perception

The structure of MFFP is presented in Fig. 5. Specifically, we first obtain the features corresponding to each scale. Next, the multiscale features are adaptively fused to generate fused features. Then, it is necessary to calculate the difference between the fused and the single-scale features. Ultimately, the output is acquired by weighting the difference features with the single-scale features. The formulaic description of the module is as follows. After an input image passes through the feature extractor and FPN, we can get a set of multiscale feature pyramids $F_s \in \{F_2, F_3, F_4, F_5\}$. However, in our method, we need to calculate the adaptive fused features Rm for each proposal box corresponding to different scale feature maps

$$F_m = \text{Conv}\left(\text{Cat}\left(F_2, F_3, F_4, F_5\right)\right) \tag{1}$$

where $F_2, F_3, F_4, F_5$, and $F_m \in \mathbb{R}^{n*c*h*w}$. n represents the count of proposal objects after filtering and c represents the channel

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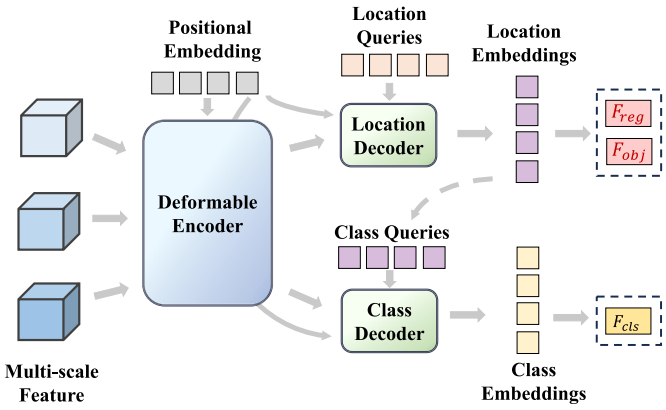



Fig. 6. Illustration of cross-layer coupling decoupling decoder. By decoupling the localization, object properties, and classification problems in multiple layers, the problem of decreased accuracy of similar known classes in the process of identifying unknown classes can be effectively alleviated.

count of the features. Subsequently, we need to compute the disparity between the fused features $F_m$ and the single-scale features $F_s$; its calculation process is as follows.

According to $F_s$ and $F_m$, we can calculate the related features $F_v$ between them. First, we need to reshape $F_s$ and $F_m$ to $F_{sr} \in \mathbb{R}^{n*c*hw}$ and $F_{mr} \in \mathbb{R}^{n*c*hw}$. Then, by multiplying $F_{sr}$ and the transpose of $F_{mr}$, the correlation matrix $V \in \mathbb{R}^{n*c*c}$ between them can be obtained. Each element in it is expressed as $v_{ij}$

$$v_{ij} = \frac{\exp\left(F_{sr}^{i} * F_{mr}^{j}\right)}{\sum_{j=1}^{c} \exp\left(F_{sr}^{i} * F_{mr}^{j}\right)}. \tag{2}$$

Also, the related feature $F_v$ between $F_s$ and $F_m$ is obtained by weighting $F_m$ with the correlation coefficient $v_{ij}$

$$R_v^i = \sum_{j=1}^{c} v_{ij} F_m^j \tag{3}$$

where $R_v^i$ denotes the related features of the ith channel. Actually, the related features are realized by channel weighting of $F_m$, and the weighting coefficients are acquired by using $F_s$. In this case, we can gain similar parts between the single-scale and fused features. Then, the difference features $F_d$ can be calculated through the difference operation of $F_m$ and $F_v$

$$F_o = F_s + \alpha\left(F_m - F_v\right). \tag{4}$$

The final output feature is obtained by weighting $F_s$, where $\alpha$ represents the weighting coefficient, and $(F_m - F_v)$ thus acts as a residual difference map, guiding the network to explicitly focus on the discrepancy between single-scale and multiscale representations. In the experiment, setting this coefficient to 0.2 can achieve better detection results.

### C. Cross-Layer Coupling Decoupling Decoder

Detection transformers employ learnable object queries to spot instances, with every query acting as a candidate object. During decoding, these queries are refined into embeddings by fusing them with the encoded feature semantics, yielding representations that jointly encode position and category for simultaneous localization and classification. In the open-world setting, the system must not only recognize previously seen categories but also pinpoint unseen regions and label them as unknown.

Mimicking human adaptation to novel scenes, we introduce a cross-layer cascade that disentangles decoding into two stages as shown in Fig. 6. A single decoder is reused: round-one embeddings isolate foreground regions and round-two embeddings assign class labels—including "unknown"—to those regions. Formally, the cascade proceeds as follows:

$$E_{\text{loc}} = \text{Decoder}\left(\text{Encoder}\left(B\left(I\right), P_n\right), Q_{\text{loc}}, R\right) \tag{5}$$

$$E_{\text{cls}} = \text{Decoder}\left(\text{Encoder}\left(B\left(I\right), P_n\right), E_{\text{loc}}, R\right) \tag{6}$$

where $B(I)$ denotes backbone features, $P_n$ is positional encoding, $Q_{\text{loc}}$ are 300 learnable location queries, and $R$ represents reference points. During cascade decoupled decoding, the very vectors that lock down spatial extent are promoted to class-specific queries; this keeps the localization step blind to category labels while still handing the identification step a map of reliable locations to guide its decision, unknown objects included.

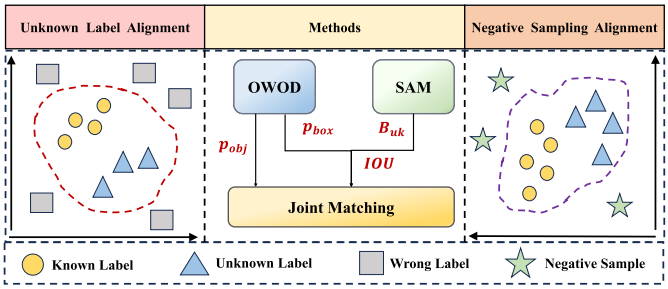



Fig. 7. Illustration of LMA. Our method matches unknown classes and background noise by predicting values and IOU correlation functions.

### D. Label Mapping Alignment

Raw outputs from SAM are highly noisy and contain both backgrounds and segmented regions. These noisy labels will significantly degrade overall performance, posing a classic noisy label learning problem. To address this issue, we propose the LMA module as shown in Fig. 7, which takes into account both objectness confidence and localization consistency to progressively learn noisy knowledge from easy to hard. These modules contain an unknown label alignment module and a negative sampling alignment.

We first use Hungarian bipartite matching to form one-to-one pairs of predictions and unknown labels $B_{\text{uk}}$. Subsequently, the unknown label alignment module utilizes a thresholding function to quantify the quality of $B_{\text{uk}}$ and filter out some of its noisy labels. This thresholding function incorporates two key variables: the objectness score $P_{\text{obj}}$ and the IOU between the prediction boxes $P_{\text{box}}$ and $B_{\text{uk}}$. $P_{\text{obj}}$ is used to identify objects, aiding in the removal of background noise. IoU is used to filter out some fragmented regions, as predicted boxes can provide an indication of dataset's annotation level, such as

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"whole," "part," and "subpart." We introduce joint selection designs to dynamically select training samples with less noise.

In this design, we combine the values of $P_{\text{obj}}$ and $\text{IoU}(P_{\text{box}(s)}, B_{\text{uk}(s)})$ to select training samples of unknown objects using a joint threshold. Specifically, we use a geometric mean function to form the thresholding function

$$s_{\text{obj}}^{\text{uk}} = \left\{ s \in S^{\text{uk}} : P_{\text{obj}}(s)^{\alpha} \,\text{IOU}\left(P_{\text{box}}(s), B_{\text{uk}}(s)^{(1-\alpha)}\right) > \tau \right\} \tag{7}$$

where $\tau$ is a selection threshold and $\alpha$ is a control hyperparameter in geometric mean. The label selection dynamically adjusts and becomes progressively accurate during the training process, which can benefit from the high recall of SAM while mitigating the impact of noisy labels.

To address the negative issue in OWOD, we also propose an SAM-based negative sampling alignment strategy, which exploits the knowledge of SAM in recognizing generic objects. Specifically, we assume that predictions with little overlap with SAM's outputs carry a high likelihood of representing backgrounds, which can effectively serve as negative samples. To support this hypothesis, we also proved the strong ability of SAM to recall unknown objects comprehensively, as stated. The pipeline of negative sampling alignment is as shown, and the set of negative objectness samples can be formalized as follows:

$$s_{\text{neg}} = \left\{ s \in S : \text{IOU}\left(P_{\text{box}}(s), B_{\text{sam}}(s)\right) > \tau_{\text{neg}} \right\} \tag{8}$$

where, in objectness samples, the objectness loss $\tau_{\text{neg}}$ is a selection threshold. After adding the negative obj, it would be modified to a loss with two terms. The inclusion of negative samples plays a crucial role in preventing the uncontrolled expansion of known class decision boundaries, which notably enhances the open-set performance.

### E. Active Learning

In the OWOD setting, a detector must absorb new categories as they appear without erasing its memory of earlier ones. To curb forgetting, the standard practice is to cache a compact image pool, or exemplars. Instead of sampling classes uniformly, we propose to mine exemplars by their objectness, an idea that costs no extra labels because we only reuse the annotations already available, unlike conventional AL that asks an oracle for more tags.

We retain objects whose objectness lies at either extreme. Low-scoring candidates are "hard" cases where the model hesitated; teaching it these cases is expected to boost recognition of future novelties, a principle embraced by modern active learning literature. High-scoring instances, conversely, are deemed prototypical; rehearsing them should stabilize old knowledge, an intuition confirmed in incremental-learning studies. After each data batch, we compute the objectness probability of every matched query embedding and keep the top and bottom 30 per class. If this yields more exemplars than prior work, we randomly downsample to match their budget, ensuring a fair comparison.

### F. Overall Loss Function

The complete training objective of our framework integrates multiple complementary loss terms to simultaneously achieve accurate detection of known objects, robust discovery of unknown objects, and effective background discrimination. The total loss is formulated as follows:

$$\begin{aligned}\mathcal{L}_{\text{tot}} = {} & \mathcal{L}_{\text{cls}} + \lambda_{\text{reg}}\mathcal{L}_{\text{reg}} + \lambda_{\text{obj}}\mathcal{L}_{\text{obj}} \\ & + \lambda_{\text{unk}}\left(\mathcal{L}_{\text{obj}}^{\text{unk}} + \lambda_{\text{reg}}\mathcal{L}_{\text{reg}}^{\text{unk}}\right) + \lambda_{\text{neg}}\mathcal{L}_{\text{obj}}^{\text{neg}}.\end{aligned} \tag{9}$$

This composite loss comprises three functional groups. The first group includes standard classification and regression losses ($\mathcal{L}_{\text{cls}}, \mathcal{L}_{\text{reg}}$), for known objects, ensuring precise localization and categorization of previously learned classes. The second group addresses objectness learning $\mathcal{L}_{\text{obj}}$ that learns a generic foreground background separation, while $\mathcal{L}_{\text{obj}}^{\text{unk}}$ and $\mathcal{L}_{\text{reg}}^{\text{unk}}$ specifically handle SAM generated unknown proposals that pass our LMA filtering, teaching the model to recognize uncertain objects as unknown and refine their localization. The third group $\mathcal{L}_{\text{obj}}^{\text{neg}}$ implements negative sampling by penalizing high objectness scores for predictions that overlap poorly with SAM masks, thereby preventing background confusion.

By jointly optimizing these terms, our framework learns a unified representation that maintains discrimination between known classes while progressively acquiring novel concepts. During incremental learning, the AL strategy interacts with this loss function by reweighting contributions from selected exemplars, amplifying signals from hard negative samples to enhance novelty detection while reinforcing prototypical positive samples to preserve established knowledge. This synergy enables stable, continuous adaptation in dynamic remote sensing environments.

## IV. Experiments

### A. Datasets

Comprehensive experiments were conducted on three challenging datasets: DIOR [22], DOTA [23], and NWPU [24] to evaluate the proposed method.

1) DIOR comprises high-resolution images from aerial and ground perspectives, featuring diverse object types such as vehicles, people, and buildings. The dataset includes 23 463 images, each sized 800 × 800 pixels, with 192 472 instances covering 20 object classes: airplane (AP), airport (AT), baseball field (BF), BC, bridge (BR), chimney (CH), dam (DA), expressway service area (ES), ET, harbor (HB), golf field (GF), ground track field (GTF), overpass (OP), ship (SH), stadium (SD), storage tank (ST), TC, TS, vehicle (VH), and windmill (WM).
2) DOTA consists of 2806 large remote sensing images ranging from 800 × 800 to 4000 × 4000 pixels, containing 188 282 instances across 16 categories: BC, baseball diamond (BD), bridge (BR), container crane (CC), GTF, harbor (HA), helicopter (HC), large vehicle (LV), plane (PL), roundabout (RA), small vehicle (SV), ship (SH), ST, soccer ball field (SBF), swimming pool (SP), and TC.
3) NWPU comprises 800 high-resolution optical remote sensing images, including 715 color images from Google

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TABLE I
COMPOSITION OF DATASETS FOR KNOWN AND UNKNOWN CLASSES

| Dataset | Setting | Known classes | unknown classes |
|---|---|---|---|
| DIOR | 16+4 | AL,AY,BF,BC,BG,CM,DM,EA,ES,GC,GF,HB,OP,SP,SD,ST | TC,TS,VH,WM |
| DIOR | 10+10 | The first 10 classes | The remaining 10 classes |
| DIOR | 4+16 | The first 4 classes | The remaining 16 classes |
| DOTA | 8+8 | BC,BD,BR,CC,GTF,HA,HC,LV | PL,RA,SV,SH,ST,SBF,SP,TC |
| NWPU | 8+2 | AP,BC,BD,BR,GTF,HB,SH,ST | TC,VH |

Earth with spatial resolutions ranging from 0.5 to 2 m, and 85 pansharpened color-infrared images from the Vaihingen dataset with a spatial resolution of 0.08 m. It contains airplanes (AP), BCs, BDs, bridges (BR), GTFs, harbors (HBs), ships (SH), STs, TCs, and vehicles (VH).

To address the open-world potential oriented object task, we adopt a known + unknown setting. During training, labels for unknown classes are omitted. In the testing phase, all instances from unknown classes are treated as unknown. Table I details the dataset splits, with DIOR adopting a $16 + 4$ setting, DOTA using $12 + 4$ setting, and NWPU with $8 + 2$ setting for comparison. To further evaluate the generalization and effectiveness of our methods, we have introduced an additional $10 + 10$ setting a $4 + 16$ setting for DIOR.

### B. Evaluation Metrics

OWOD is often evaluated in terms of two aspects, recall on unknown objects and precision on known objects. Metrics of precision include mean average precision (mAP), wilderness impact (WI), and absolute open-set error (AOSE). To evaluate the closed-set performance, Known mAP is used and further partitioned into previously and newly introduced object classes. To evaluate the open-set performance, WI and AOSE are used to quantify the errors in recognizing unknown objects as known objects. WI is a metric that measures relative error in classifying unknown objects as known classes, formulated as follows:

$$\mathrm{WI} = \frac{P_K}{P_K \cup U} - 1 \tag{10}$$

where $P_K$ is the precision of known classes and $P_K \cup U$ is the precision of both known classes and unknown classes. AOSE measures the absolute error, which is the number of recognizing unknown objects as known objects. The metric of unknown object recall is used to quantify the comprehensiveness of unknown object detection.

### C. Implementation Details

The default detector for our method is based on the D-DETR model [11], utilizing a DINO-pretrained ResNet-50 FPN backbone. Multiscale features are first extracted; then, learnable queries are fed to the deformable decoder to yield query embeddings. Each embedding proposes one region, and the 300 highest scoring boxes per image are kept for testing. Training is distributed across two RTX-3090 24GB GPUs.

TABLE II
DETECTION RESULTS OF DIFFERENT METHODS ON DIOR UNDER $16 + 4$ SETTING

| Method | 16 | | 16-4 | | |
|---|---|---|---|---|---|
| | U-Recall | mAP | PK | CK | mAP |
| ORE | 14.18 | 50.62 | 46.30 | 67.11 | 50.46 |
| OW-DETR | 18.53 | 51.58 | 49.71 | 67.46 | 53.26 |
| PROB | 23.37 | 53.90 | 51.61 | 71.01 | 55.49 |
| KTCN | 27.59 | 55.88 | 52.96 | 70.90 | 56.55 |
| SGROD | 35.82 | 55.68 | 52.22 | 71.22 | 56.02 |
| Ours | **42.93** | **57.69** | **54.23** | **75.47** | **58.48** |

We employ the Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.99$, and a weight decay of $10^{-4}$. We start with a learning rate of $3 \times 10^{-3}$ and divide it by 10 once epoch 30 is reached; in the incremental phase we reset it to $4 \times 10^{-4}$. For rehearsal, we reserve 50 exemplars for every previously seen class. The coefficients for the unknown object loss are identical to $L_{\mathrm{reg}}$ and $L_{\mathrm{obj}}$. We sample a $32 \times 32$ grid in SAM's "everything" regime and retain masks whose forecast IoU exceeds 0.9 and stability surpasses 0.85.

### D. Quantitative Analysis

In this section, to elucidate the operational mechanism of our approach, we conducted a series of experiments on DIOR, DOTA, and NWPU datasets as shown next. The detection performance for each known class and unknown class is demonstrated through mAP, uAP. This rigorous experimental analysis is vital to evaluate the efficacy of our method when pitted against other established and advanced techniques in the field.

*1) DIOR:* We contrast our method with other approaches on DIOR using $16+4$ and $10+10$ settings. Table II compares the results of our method with other related methods on the DIOR dataset. Based on the experimental results, although the baseline method achieves some improvement in detecting unknown classes outperforming all previous approaches, it still exhibits suboptimal performance. In contrast, our proposed method not only surpasses prior methods in open-set detection but also maintains a competitive detection accuracy. Under the first split setting, the proposed method improves the recall rate for unknown classes by 7.09% and increases the detection

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TABLE III
DETECTION RESULTS OF DIFFERENT METHODS ON DIOR UNDER 10 + 10 SETTING

| Method | 10 | | 10-10 | | |
|---|---|---|---|---|---|
| | U-Recall | mAP | PK | CK | mAP |
| ORE | 21.41 | 54.62 | 49.61 | 53.11 | 51.36 |
| OW-DETR | 28.01 | 56.58 | 49.54 | 55.20 | 52.37 |
| PROB | 27.77 | 57.90 | 54.21 | 55.01 | 54.61 |
| KTCN | 33.10 | 59.22 | 56.12 | 58.12 | 57.12 |
| SGROD | 39.25 | 59.17 | 55.02 | 61.11 | 58.07 |
| Ours | **46.33** | **62.51** | **57.09** | **63.21** | **60.15** |

TABLE IV
DETECTION RESULTS OF DIFFERENT METHODS ON DIOR UNDER 4 + 16 SETTING

| Method | 4 | | 4-16 | | |
|---|---|---|---|---|---|
| | U-Recall | mAP | PK | CK | mAP |
| ORE | 16.77 | 58.90 | 59.04 | 53.66 | 54.74 |
| OW-DETR | 21.08 | 64.62 | 62.44 | 54.60 | 56.17 |
| PROB | 19.57 | 67.10 | 61.91 | 51.17 | 53.32 |
| KTCN | 23.67 | 65.34 | 59.76 | 57.89 | 58.23 |
| SGROD | 31.77 | **69.18** | 63.17 | 56.92 | 58.17 |
| Ours | **41.61** | 68.79 | **67.39** | **59.26** | **65.76** |

TABLE V
DETECTION RESULTS OF DIFFERENT METHODS ON DOTA UNDER 8 + 8 SETTING

| Method | 8 | | 8-8 | | |
|---|---|---|---|---|---|
| | U-Recall | mAP | PK | CK | mAP |
| ORE | 15.01 | 41.92 | 39.72 | 56.11 | 47.92 |
| OW-DETR | 13.79 | 43.55 | 39.23 | 57.01 | 48.12 |
| PROB | 19.05 | 43.09 | 40.06 | 59.01 | 49.54 |
| KTCN | 25.02 | 46.80 | 44.99 | 58.60 | 51.80 |
| SGROD | 31.90 | 45.11 | 43.69 | 59.96 | 51.82 |
| Ours | **36.44** | **47.99** | **45.84** | **61.24** | **53.54** |

TABLE VI
DETECTION RESULTS OF DIFFERENT METHODS ON NWPU UNDER 8 + 2 SETTING

| Method | 8 | | 8-2 | | |
|---|---|---|---|---|---|
| | U-Recall | mAP | PK | CK | mAP |
| ORE | 27.18 | 83.19 | 81.30 | 81.40 | 81.32 |
| OW-DETR | 26.52 | 83.58 | 82.15 | 78.70 | 81.46 |
| PROB | 38.49 | 84.16 | 84.21 | 78.01 | 82.97 |
| KTCN | 36.74 | 83.78 | **86.50** | 80.90 | 85.38 |
| SGROD | 46.82 | 82.68 | 83.72 | 81.02 | 83.18 |
| Ours | **55.90** | **86.03** | 86.23 | **85.73** | **86.13** |

accuracy for known objects by 2.01%. Furthermore, after all unknown objects are annotated and the model continues detection, the precision improves by 2.46% compared to previous methods.

Table III provides the detailed results of the detection performance of our method for each known and unknown class under the 10 + 10 setting. The results demonstrate that even with a limited number of classes designated as known, our method is capable of detecting potential objects effectively. Moreover, our method significantly outperforms baseline methods in terms of uR of 7.08%, mAP of 2.07%, and 2.08% after incremental learning, demonstrating the unique advantages of our approach.

To evaluate our method under the challenging scenario where unknown categories drastically outnumber known ones, we created a DIOR 4 + 16 split. Results in Table IV demonstrate that our method achieves 41.61% U-Recall, 9.84% higher than SGROD (31.77%). Despite the extreme class imbalance, known class mAP remains robust at 68.79%, proving that CCDD effectively isolates known class learning from massive unknown noise. Moreover, our PK (67.39%) and CK (59.26%) both exceed baselines, showing superior knowledge retention when learning 16 new classes sequentially.

*2) DOTA:* Table V presents the results of our method on the DOTA dataset using the 8+8 setting. As observed from the table, our approach not only effectively detects known classes but also demonstrates strong detection performance for potential oriented objects of unknown classes. Compared to baseline methods that require more prior knowledge to finely classify unknown objects, our method significantly outperforms them in terms of unknown object detection. Although our method cannot perform fine-grained classification of unknown classes, its detection performance and unique capability to detect objects in remote sensing images make it an indispensable approach. In addition, in this set of experiments, we integrated our proposed method into the DETR framework. The experimental results clearly demonstrate that our method consistently enhances the model's ability to detect unknown object classes across different frameworks.

*3) NWPU:* NWPU is a specialized dataset, where the interclass differences between different categories are minimal. Table VI presents the performance metrics on the NWPU test set after adopting the 8 + 2 setting. From the table results, although our method slightly reduces mAP, there is a significant improvement in the detection performance of potential objects from unknown categories. Similar to Table V, our method's detection performance for unknown objects is significantly higher than that of existing other detection methods.

### E. Ablation Study

To better understand how the proposed method works, we conduct a series of ablation studies on the DIOR dataset using the 16 + 4 setting.

*1) Effectiveness of Each Cpmponent:* The experimental results of various methods are shown in Table VII. Training the model using the baseline method yields a mAP of 54.25% and an unknown recall rate of 36.05%. After introducing the CCDD structure, the recall rate for unknown classes increases

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TABLE VII
EFFECTIVENESS OF EACH COMPONENT OF THE PROPOSED METHOD

| Component | | | | 16 | | 16-4 | | |
|---|---|---|---|---|---|---|---|---|
| MFFP | CCDD | LMA | AL | U-Recall | mAP | PK | CK | mAP |
| ✓ | | | | 36.05 | 54.25 | 53.27 | 72.38 | 57.09 |
| ✓ | ✓ | | | 38.71 | 56.20 | 42.85 | **78.21** | 49.92 |
| ✓ | ✓ | ✓ | | 42.93 | 57.69 | 45.46 | 78.01 | 52.01 |
| ✓ | ✓ | | ✓ | 38.71 | 56.20 | **56.30** | 68.23 | **58.69** |
| ✓ | ✓ | ✓ | ✓ | **42.93** | **57.69** | 54.23 | 75.47 | 58.48 |

TABLE VIII
EFFECTIVENESS OF DIFFERENT ORDERS OF THE PROPOSED METHOD

| Method | 16 | | 16-4 | | |
|---|---|---|---|---|---|
| | U-Recall | mAP | PK | CK | mAP |
| SGROD1 | 35.82 | 55.68 | 52.22 | 71.22 | 56.02 |
| Ours1 | **42.93** | **57.69** | **54.23** | **75.47** | **58.48** |
| SGROD2 | 33.51 | 54.78 | 54.62 | 65.87 | 56.87 |
| Ours2 | **43.79** | **55.19** | **55.53** | **72.58** | **58.94** |
| SGROD3 | 31.31 | 57.21 | 54.12 | **57.12** | 54.72 |
| Ours3 | **35.29** | **60.54** | **59.93** | 55.48 | **59.04** |

TABLE IX
EFFECTIVENESS OF SAM CONFIDENCE OF THE PROPOSED METHOD

| Method | 16 | | 16-4 | | |
|---|---|---|---|---|---|
| | U-Recall | mAP | PK | CK | mAP |
| 0.6 | 39.86 | 55.29 | 54.30 | 75.14 | 58.47 |
| 0.65 | 40.23 | 56.10 | 51.61 | 77.42 | 56.77 |
| 0.7 | 40.71 | 55.33 | 53.19 | 77.47 | 58.04 |
| 0.75 | 40.09 | 58.23 | 55.16 | 72.12 | 58.55 |
| 0.8 | 41.46 | 58.20 | 54.19 | 74.34 | 58.22 |
| 0.85 | 42.49 | 57.78 | 53.96 | 76.39 | 58.45 |
| 0.9 | 42.93 | 57.69 | 54.23 | 75.47 | 58.48 |
| 0.95 | 42.52 | 57.08 | 54.06 | 74.92 | 58.23 |

to 38.71%, demonstrating the effectiveness of separating localization and classification in open-set detection. Furthermore, when the CCDD module is used in combination with either the LMA module or both modules together, the proposed method achieves improvements of 4.28% in U-Recall and 1.49% in mAP over the baseline. Ultimately, the best result is an mAP of 58.69% without compromising detection performance. This outcome highlights the effectiveness and applicability of the proposed approach in continuous open-world learning tasks.

The ablation results validate the distinct roles of our proposed modules. Specifically, the introduction of the MFFP module leads to a substantial gain in U-Recall. This confirms that fusing multiscale features is critical for capturing remote sensing objects with extreme scale variations. Furthermore, the CCDD module contributes significantly to the mAP, indicating that decoupling the objectness and classification heads effectively resolves the optimization conflict between known and unknown classes.

The strong performance of each individual module indicates that incrementally enhancing the model's sensitivity to unknown categories through a modular design can lead to significant improvements in open-set detection. By incorporating multiple strategic components, the model can more robustly recognize and adapt to dynamically evolving category sets, demonstrating exceptional performance, especially in scenarios involving class imbalance and category expansion.

*2) Effectiveness of Different Order:* To evaluate the impact of different class orders on OWOD learning methods, experiments were conducted on the DIOR dataset. Specifically, four random class orders were generated for the 16 object categories in the DIOR dataset. The experimental results are shown in Table VIII. Among them, Order 4 yielded the best performance, achieving a U-Recall of 43.79%, which is 10.28% higher than the baseline method. For the other class orders, our proposed method also outperformed the baseline by at least 3.96% in recall. These results demonstrate that the proposed method is effective in the continual open-world detection setting, regardless of the order in which categories are introduced.

*3) Effectiveness of SAM Confidence:* To evaluate the impact of different class orders on OWOD learning methods, experiments were conducted on the DIOR dataset. In this experiment, we set multiple threshold values for segmentation and recorded key metrics such as precision and recall under each threshold. These metrics allow us to observe how the model's accuracy varies with different threshold settings. As the data demonstrated in Table IX, varying the confidence threshold from 0.6 to 0.95 results in only minor fluctuations across all key performance metrics. This analysis confirms that our proposed method is not sensitive to the precise setting of this parameter and maintains stable performance, which underscores its robustness and practical applicability.

*4) Effectiveness of VLM:* As shown in Table X, we included the experimental results of Grounding DINO, a popular large-scale vision-language model. We used textual prompts describing the objects to query Grounding DINO and obtain the predicted bounding boxes. The results clearly demonstrate that the performance of using SAM outperforms that of Grounding DINO, validating our choice of SAM in this work. Furthermore, introducing Grounding DINO into the baseline model improved U-Recall by 9.33%, indicating the advantage of leveraging the rich prior knowledge embedded in

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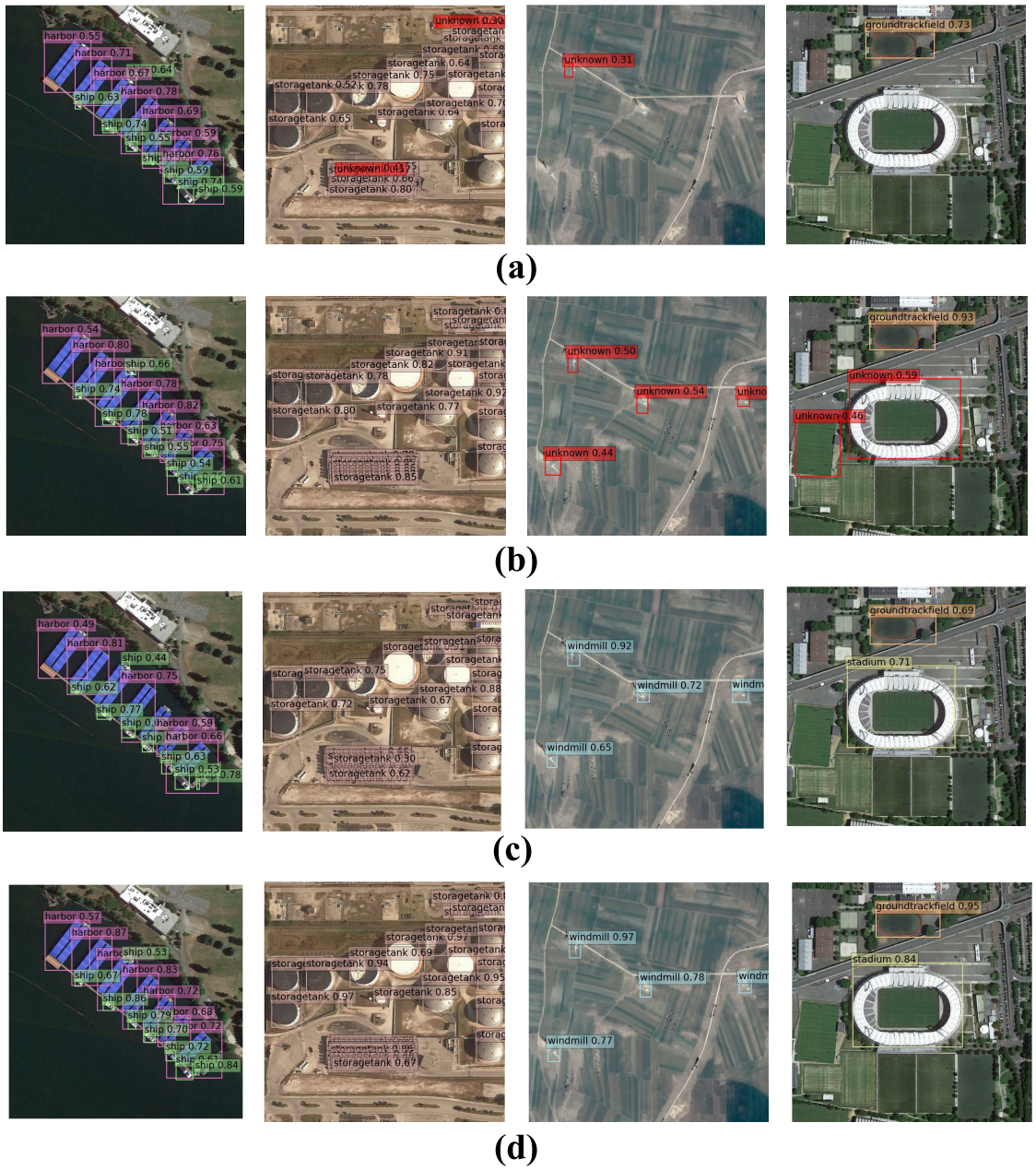



Fig. 8. Visualization detection results of DIOR dataset. (a) SGROD method in Step 1. (b) Our method in Step 1. (c) SGROD method in Step 2. (d) Our method in Step 2. Unknown classes are marked by red boxes.

TABLE X
EFFECTIVENESS OF VLM OF THE PROPOSED METHOD

| Method | 16 | | 16-4 | | |
|---|---|---|---|---|---|
| | U-Recall | mAP | PK | CK | mAP |
| GroundingDINO | 33.49 | 55.46 | 53.75 | **76.39** | 58.28 |
| SAM | **42.93** | **57.69** | **54.23** | 75.47 | **58.48** |
| Qwen-max | 11.49 | 16.62 | 33.75 | 26.39 | 32.28 |
| GPT-4o | 14.79 | 21.69 | 35.17 | 24.11 | 32.96 |
| Gemini-2.5-pro | 28.96 | 34.79 | 41.01 | 36.03 | 40.01 |

vision-language models. Importantly, these results also suggest that our method is not limited to SAM and can be effectively extended to other vision-language models as well.

To quantify the ceiling of SAM against modern closed-source large models, we evaluated Qwen-max, GPT-4o, and Gemini-2.5-Pro using their vision APIs for zero-shot object detection on DIOR $16+4$. As shown in the expanded table below, SAM significantly outperforms both large models. Despite closed-source large models' general visual prowess, its training bias toward natural images and language grounding limits performance on aerial imagery. SAM's pure segmentation focus provides superior localization priors for unknown object discovery, validating our design choice.

*5) Effectiveness of AL:* To optimize our AL strategy, we investigated the impact of different sample selection criteria on model performance, with the detailed results presented in Table XI. The findings reveal a crucial insight: strategies

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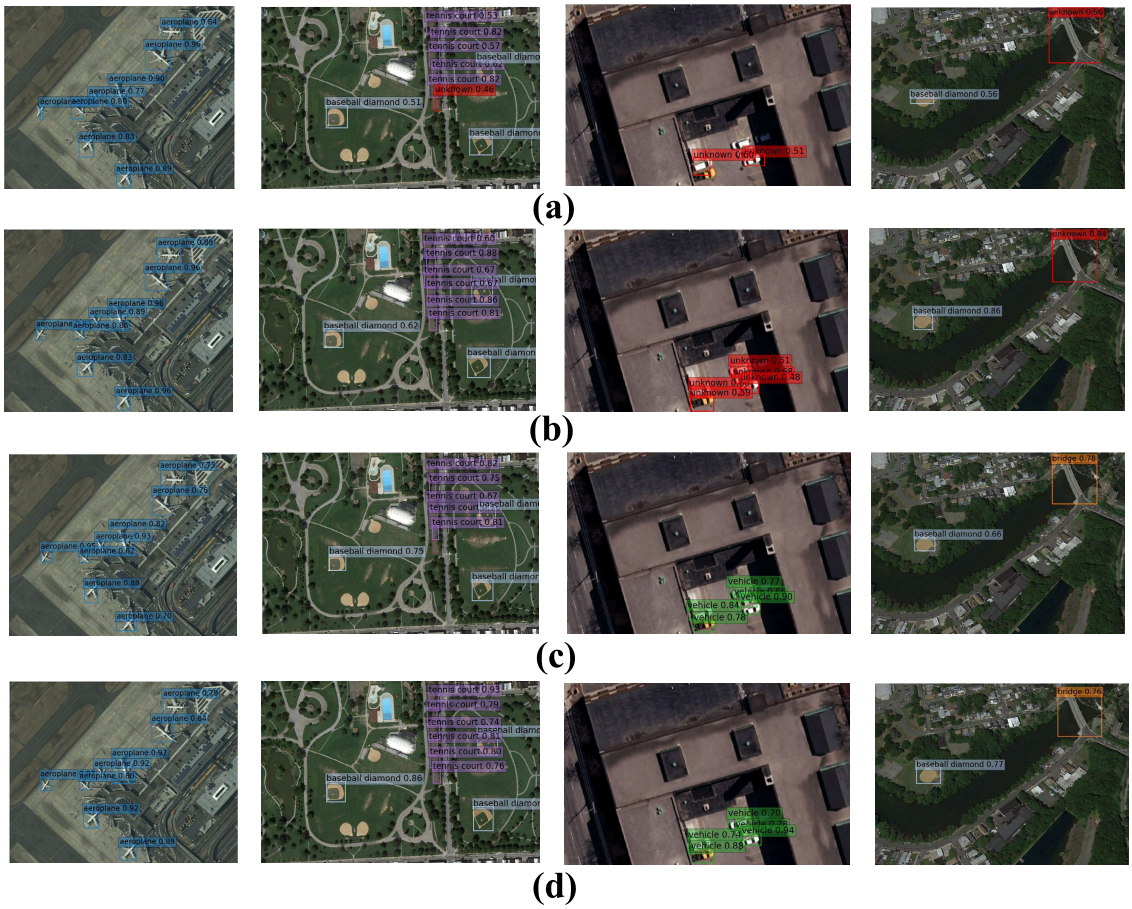



Fig. 9. Visualization detection results of NWPU dataset. (a) SGROD method in Step 1. (b) Our method in Step 1. (c) SGROD method in Step 2. (d) Our method in Step 2. Unknown classes are marked by red boxes.

TABLE XI
EFFECTIVENESS OF AL

| Hardest/Easiest | 16 | | 16-4 | | |
|---|---|---|---|---|---|
| | U-Recall | mAP | PK | CK | mAP |
| 0/60 | 41.46 | 58.20 | **57.20** | 52.03 | 56.17 |
| 30/30 | 41.46 | 58.20 | 54.19 | 74.34 | 58.22 |
| 50/50 | 41.46 | 58.20 | 53.92 | 78.07 | **58.75** |
| 60/0 | 41.46 | 58.20 | 42.71 | **78.88** | 49.94 |

relying on extreme ends of the difficulty spectrum such as selecting exclusively the most uncertain samples or the least uncertain samples, both lead to suboptimal performance, particularly in discovering new classes. In contrast, a balanced strategy that selects a mix of samples with varying complexity yields significantly better results. We hypothesize that this is because learning from a diverse set of samples prevents the model from overfitting to specific error types and fosters the development of more robust and generalizable features. Furthermore, the performance difference between the 30/30 and 50/50 splits is marginal, suggesting that our method is not highly sensitive to the exact ratio, so long as a balanced sampling approach is adopted.

### F. Qualitative Analysis

*1) Experiments on the DIOR Dataset:* In the visualization comparison of detection results on the DIOR dataset, as shown in Fig. 8, we present the performance differences between the baseline method and our proposed method in open-set recognition and incremental detection tasks. The first row in the figure represents the detection results of the baseline method, while the second row shows the results of our proposed method.

By comparison, it is evident that our method demonstrates a clear advantage in detecting unknown objects, achieving higher recall and providing more accurate identification of unknown-category targets. In contrast, the baseline method exhibits a higher number of missed detections and fails to effectively recognize newly emerging categories. This observation indicates that our proposed method performs significantly better in open-set recognition tasks, effectively reducing the miss rate for unknown objects. OWOD has emerged as a more demanding and realistic task.

Moreover, our method also demonstrates excellent performance in recognizing known categories. Compared to the baseline, the proposed approach is less likely to misclassify known categories as unknown, resulting in more accurate and reliable detection outcomes. However, in the final detection example, although our method successfully identifies objects not present in the training dataset, a few instances of

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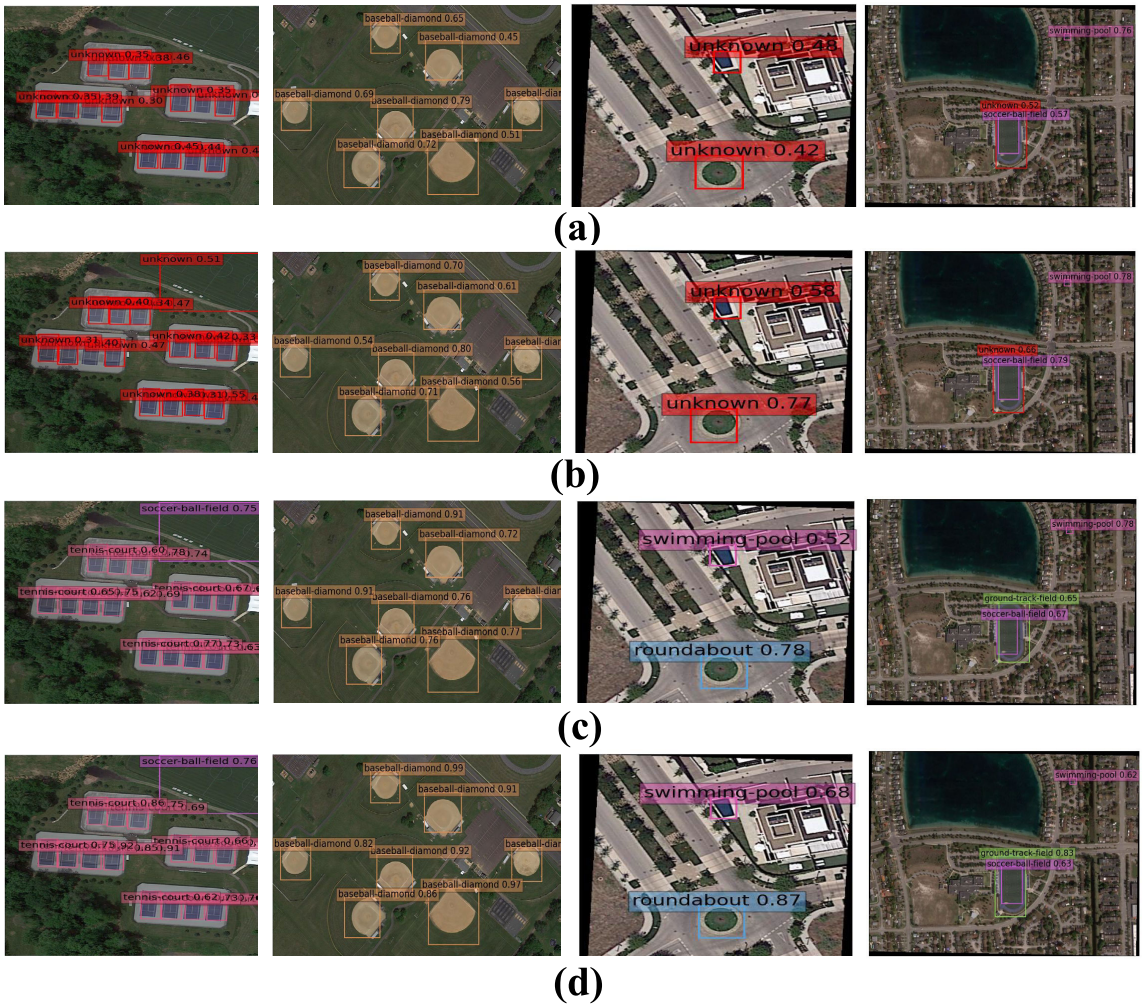



Fig. 10. Visualization detection results of DOTA dataset. (a) SGROD method in Step 1. (b) Our method in Step 1. (c) SGROD method in Step 2. (d) Our method in Step 2. Unknown classes are marked by red boxes.

background being mistakenly recognized as unknown objects were observed. This indicates that while our method significantly improves detection accuracy in open-set recognition tasks, further optimization is still needed in specific background scenarios to reduce the false detection rate of background regions.

In the incremental detection task, we further analyzed the DIOR dataset to validate the performance stability of the proposed method in incremental learning. During the incremental detection process, the proposed method effectively mitigates the issue of catastrophic forgetting, meaning that the detection performance on previously learned classes is not adversely affected by the training of new classes.

*2) Experiments on the NWPU Dataset:* In the visualization comparison of detection results on the NWPU dataset, as shown in Fig. 9, we illustrate the performance differences between the baseline method and our proposed method in the open-set recognition task. Our method achieves higher recall when detecting unknown objects, enabling more accurate identification of unknown category targets. Notably, in the third image, the proposed method successfully detects vehicle, an unknown category, while the baseline method exhibits a large number of missed detections and fails to effectively capture the new category. This observation demonstrates that our proposed method is better equipped to handle dynamic category changes in open-set recognition, showing a clearer and more robust recall performance.

To gain deeper insight into how the two strategies behave during incremental learning, we carried out a qualitative inspection on the NWPU dataset. Fig. 9 displays representative image crops for side-by-side comparison. The upper row visualizes the baseline detector's outputs, while the lower row illustrates the corresponding predictions produced by our approach.

The proposed method demonstrates superior incremental recognition capability for unknown categories. In the baseline method, the model exhibits misidentification issues when detecting unknown objects, and even during subsequent incremental learning stages, it fails to improve detection performance for those categories. In contrast, the proposed method shows stronger adaptability in incremental learning, consistently recognizing and learning new object categories.

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This clearly highlights the effectiveness and importance of our method in identifying unknown targets during incremental detection.

*3) Experiments on the DOTA Dataset:* Following the same experimental protocol, we extended our evaluation to the large-scale DOTA dataset to further verify the robustness of our method. As illustrated in the visual comparisons in Fig. 10, the results are highly consistent with the findings from the DIOR and NWPU datasets. Our proposed method again demonstrates excellent results in both open-set recognition and continual learning scenarios. Particularly noteworthy is the method's enhanced perceptual capability for unknown classes, successfully identifying a wider range of novel objects that were missed by the baseline. Furthermore, the model maintained high accuracy on previously learned categories after incrementally learning new ones, confirming its strong ability to mitigate catastrophic forgetting. These consistent results across multiple diverse datasets robustly validate the effectiveness and generalizability of our approach.

Overall, the proposed method significantly outperforms the baseline on three datasets, not only improving the recall and recognition accuracy for unknown categories but also demonstrating higher detection confidence, especially in open-set and incremental learning scenarios. Compared to the baseline, the proposed method shows a clear advantage in new category detection tasks, enhancing the ability to detect new targets while better preserving the performance on previously learned classes, effectively mitigating catastrophic forgetting during incremental learning.

## V. Conclusion

In order to enhance the generalizability and practicality of object detection methods in open-world remote sensing scenarios, this article proposes an SAM-guided OWOD framework that integrates four key modules: MFFP, CCDD, LMA, and AL. Experiments on DIOR, DOTA, and NWPU datasets validate the effectiveness of our approach, demonstrating significant improvements in unknown object recall while maintaining robust known-class detection.

Our method achieves superior unknown class discovery compared to existing methods, effectively addressing RS-specific challenges like extreme scale variation and subtle interclass differences. However, the primary limitation is the additional computational cost of SAM inference, and performance is sensitive to SAM hyperparameter thresholds, requiring careful tuning for new datasets. Future research will focus on enhancing feature representation for dense small objects within unknown categories and exploring unsupervised domain adaptation to minimize manual parameter tuning in dynamic environments.

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